ISyE6669 Deterministic Optimization
Homework 12
Fall 2025

Note: In all your answers, you need to give sufficient details and explanation how you derive or come to your solution.

1. In the lectures, we introduced the concept of a convex cone and gave three important examples: the nonnegative orthant cone $\mathbb{R}^n_+$, the second-order cone (SOC) $\mathbb{L}^n$, and the positive semidefinite cone (PSD) $\mathbb{S}^n_+$. Given a convex cone K, we can compare vectors using the cone K as $x \succeq_K y$ if and only if $x - y \in K$.

   For example, if the cone is the nonnegative orthant $\mathbb{R}^n_+ = \{(x_1, \dots, x_n) : x_i \geq 0, \ \forall i = 1, 2, \dots, n\}$. Then comparing vectors with respect to $\mathbb{R}^n_+$ is the familiar componentwise comparison. That is, $x \succeq_{\mathbb{R}^n_+} y$ if and only if $x_i \geq y_i$ for each $i = 1, 2, \dots, n$. If the cone is $\mathbb{L}^n$ or $\mathbb{S}^n_+$, the comparison is not componentwise comparison anymore.

   As an exercise, let

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad y = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}.$$

   Answer the following questions.

   (a) Using $\mathbb{R}^3_+$, is it true that $x \succeq_{\mathbb{R}^3_+} y$? Explain why. Plot $x - y$ and $\mathbb{R}^3_+$.

   Solution: If the cone is the nonnegative orthant $\mathbb{R}^n_+ = \{(x_1, \dots, x_n) : x_i \geq 0, \ \forall i = 1, 2, \dots, n\}$, then comparing vectors with respect to $\mathbb{R}^n_+$ is the familiar componentwise comparison. In this case $\mathbb{R}^3_+ = \{(x_1, x_2, x_3) : x_i \geq 0, \ \forall i = 1, 2, 3\}$. After comparing vectors we see that:

$$x - y = \begin{bmatrix} 1-3 \\ 2-2 \\ 3-2 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix}.$$

   Since first component is negative $x - y \notin \mathbb{R}^3_+$ so it's NOT TRUE that $x \succeq_{\mathbb{R}^3_+} y$. The plot from Figure 1 also shows that $x - y$ is outside the $\mathbb{R}^3_+$ cone.

   (b) Using $\mathbb{L}^3$, is it true that $x \succeq_{\mathbb{L}^3} y$? Explain why. Plot $x - y$ and $\mathbb{L}^3$.

   Solution: Based on the lecture material:

$$x \succeq_{\mathbb{L}^3} y \quad \Leftrightarrow \quad x - y \succeq_{\mathbb{L}^3} 0 \quad \Leftrightarrow \quad x - y \in L^m$$

   where:

$$\mathbb{L}^3 = \{(x, y, z) : \sqrt{x^2 + y^2} \leq z\} = \{(x, y, z) : ||[x; y]||_2 \leq z\}$$

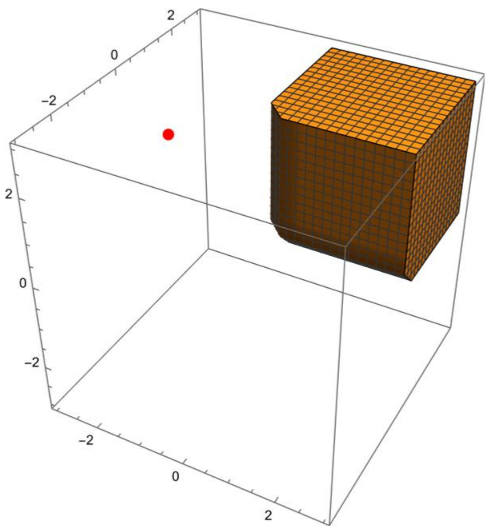

Figure 1: $x - y$ and $\mathbb{R}^3_+$.

Since $x - y = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix} \in L^m$? and $\sqrt{(-2)^2 + 0^2} \leq 2^2$ we have $x - y \in \mathbb{L}^3$ and thus it is TRUE that $x \succeq_{\mathbb{L}^3} y$

The plot in Figure 2 also shows that $x - y$ is inside the $\mathbb{L}^3$ cone.

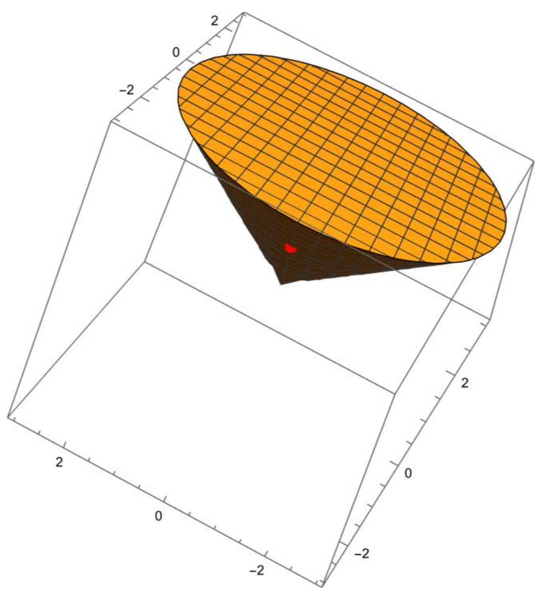

Figure 2: $x - y$ and $\mathbb{L}^3$.

(c) Now define three matrices

$$A = \begin{bmatrix} -6 & 7 & 8 \\ 7 & -8 & 9 \\ 8 & 9 & -10 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} -10 & 9 & 8 \\ 9 & -8 & 7 \\ 8 & 7 & -12 \end{bmatrix}.$$

Is it true that $A \preceq_{\mathbb{S}^3_+} B$? Is it true that $A \succeq_{\mathbb{S}^3_+} C$? Explain why. Remember A real symmetric matrix is positive semidefinite if and only if all of its eigenvalues are nonnegative.

Solution: from the lectures we know that $A \preceq_{\mathbb{S}^3_+} B$ is equivalent to $B - A \in \mathbb{S}^3_+$

$$B - A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} - \begin{bmatrix} -6 & 7 & 8 \\ 7 & -8 & 9 \\ 8 & 9 & -10 \end{bmatrix} = \begin{bmatrix} 6 & -6 & -6 \\ -6 & 10 & -6 \\ -6 & -6 & 14 \end{bmatrix}$$

Matrix $B - A$ is not psd because its third principal minor

$$M_3^{(B-A)} = \det\left(\begin{bmatrix} 6 & -6 & -6 \\ -6 & 10 & -6 \\ -6 & -6 & 14 \end{bmatrix}\right) = -672$$

is negative. Alternatively, you can show that matrix $B - A$ is not psd because one of its eigenvalues $\lambda = 4(2 - \sqrt{7})$ is negative. Therefore it is NOT TRUE that $A \preceq_{\mathbb{S}^3_+} B$.

Similar as before, $A \succeq_{\mathbb{S}^3_+} C$ is equuivalent to $A - C \in \mathbb{S}^3_+$

$$A - C = \begin{bmatrix} -6 & 7 & 8 \\ 7 & -8 & 9 \\ 8 & 9 & -10 \end{bmatrix} - \begin{bmatrix} -10 & 9 & 8 \\ 9 & -8 & 7 \\ 8 & 7 & -12 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & 2 \end{bmatrix}$$

This matrix is not psd because its second principal minor

$$M_2^{(A-C)} = \det\left(\begin{bmatrix} 4 & -2 \\ -2 & 0 \end{bmatrix}\right) = -4$$

is negative. Alternatively, you can show that matrix $A - C$ is not psd because one of its eigenvalues $\lambda \approx -1.76$ is negative. Thus $A - C \notin \mathbb{S}^3_+$ and hence it is NOT TRUE that $A \succeq_{\mathbb{S}^3_+} C$.

2. There are n villages with known coordinates $v_1 = (x_1, y_1), v_2 = (x_2, y_2), \ldots, v_n = (x_n, y_n)$, where x_i is the x-coordinate of the point v_i and y_i is the y-coordinate of the point v_i.

   We want to locate a fire station $v = (x, y)$ such that the longest distance from the fire station to a possible fire in the villages is minimized. The distance is measured by the Euclidean distance. That is, the distance between two points $u = (a_1, b_1)$ and $w = (a_2, b_2)$ is given by $\|u - w\| = \sqrt{(a_1 - a_2)^2 + (b_1 - b_2)^2}$. Here, $\|u - w\|$ is also called the ℓ_2 norm of the vector $u - w$.

Formulate a second-order conic program (SOCP) for this problem. Note that the objective function of your SOCP must be a linear function of the variables and the constraints must be either linear constraints or second-order conic constraints.

Solution: Distance from village $v_i = (x_i, y_i)$ to fire station $v = (x, y)$ is given by $\|v - v_i\| = \sqrt{(x - x_i)^2 + (y - y_i)^2}$, hence longest distance between fire station and possible fire is given by:

$$\max_{i=1,\ldots,n} \{\|v - v_i\|\} = \max_{i=1,\ldots,n} \{\sqrt{(x - x_i)^2 + (y - y_i)^2}\}.$$

Thus problem can be formulated as:

$$\min \max_{i=1,\ldots,n} \{\|v - v_i\|\} = \min \max_{i=1,\ldots,n} \{\sqrt{(x - x_i)^2 + (y - y_i)^2}\}$$

Above problem can be reformulated as SOCP program:

$$\begin{array}{ll} \min & z \\ \text{s.t.} & \sqrt{(x - x_i)^2 + (y - y_i)^2} \leq z, \ \forall i = 1, \ldots, n \end{array}$$

Randomly generate the coordinates of $n = 20$ villages: generate the x-y coordinate of a village as a pair of random numbers uniformly distributed on the interval $[0, 10] \times [0, 10]$. Implement your SOCP model in CVXPY using the coordinates of the 20 villages that you generate. Solve the model and output the fire station's location and the minimum longest distance from the first station to a village.

Solution: For 20 villages located at:

$$\begin{aligned}
v_1 &= (6.05978279,\ 7.33369361) \\
v_2 &= (1.38947157,\ 3.12673084) \\
v_3 &= (9.97243281,\ 1.28162375) \\
v_4 &= (1.78993106,\ 7.52925429) \\
v_5 &= (6.62160514,\ 7.84310132) \\
v_6 &= (0.96894396,\ 0.58571285) \\
v_7 &= (9.6239599,\ 6.16557444) \\
v_8 &= (0.86629961,\ 5.61272363) \\
v_9 &= (6.16524709,\ 9.63843023) \\
v_{10} &= (5.74304294,\ 3.71160848)
\end{aligned}$$

$$v_{11} = (4.52145244,\ 2.01850248)$$
$$v_{12} = (5.69305118,\ 1.95095974)$$
$$v_{13} = (5.83704016,\ 4.76313474)$$
$$v_{14} = (5.178144,\ 8.23098634)$$
$$v_{15} = (7.32225027,\ 0.69056275)$$
$$v_{16} = (6.72128935,\ 6.43484806)$$
$$v_{17} = (8.2801437,\ 2.04469394)$$
$$v_{18} = (6.17488953,\ 6.17701012)$$
$$v_{19} = (3.01068621,\ 8.71740586)$$
$$v_{20} = (5.89654082,\ 9.81770093)$$

best location for the fire station is $v = (5.2142959,\ 4.25079822)$. Maximum distance from fire station to given villages is 5.608552739931096.

Code used is shown bellow:

```
import numpy as np
import cvxpy as cp

# generate village locations
rnd = np.random.RandomState(2021)
vill = rnd.uniform(0, 10, size = (20, 2))
print("Coordiantes of the villages are: ", vill)

# set variable whos first two coordinates are location of the fire station,
# and third one is its maximum distance from villages
v = cp.Variable(3)
# set the problem - objective and constraints
obj = cp.Minimize(v[2])
# setting constraint using norm
#const = [cp.norm(vill[i] - v[:2], 2) <= v[2] for i in range(20)]
#setting constraints using SOC
const = [cp.SOC(v[2], vill[i] - v[:2]) for i in range(20)]
prob = cp.Problem(obj, const)
#solve problem
prob.solve()

print("Coordinates of the fire station are:", v[:2].value)
print("Largest distance from fire station to potential fire is: ", v[2].value)
```